

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES



ITEP 414

System Administration and Maintenance

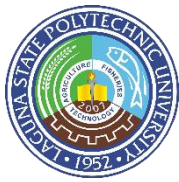
Week 3 Portfolio Project

Enterprise Server Deployment and Operating System Installation

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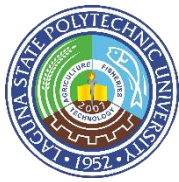
BSIT 4C

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BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is traditional firmware that initializes hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is the modern firmware interface designed to replace legacy BIOS.
Boot Process	Performs POST, looks for a bootable device, reads boot information from the MBR, and starts the bootloader.	Initializes hardware and loads a UEFI bootloader from the EFI System Partition. It can also provide a more flexible boot manager. (UEFI Forum)
Maximum Disk Support	Traditionally limited by the MBR partitioning scheme to about 2 TB for standard 512-byte sectors.	Works with GPT and supports disks much larger than 2 TB. GPT theoretically supports capacities up to about 9.4 ZB under the commonly referenced 512-byte-sector calculation. (Microsoft Learn)
Partition Style	Mainly uses MBR (Master Boot Record).	Commonly uses GPT (GUID Partition Table). GPT supports many more partitions and maintains backup partition information. (UEFI Forum)
Security Features	Has limited built-in boot security compared with modern UEFI systems.	Supports Secure Boot, which can verify digitally signed boot software before allowing it to run. (Microsoft Support)
Speed	Generally, has a slower and more limited startup process.	Generally, provides faster boot and resume times because of its newer firmware architecture. (Microsoft Learn)
Advantages	Simple, widely compatible with older hardware, and useful for legacy systems.	Faster startup, better security, support for large drives, GPT, more partitions, and a more flexible firmware environment.
Disadvantages	Older technology, limited disk/partition support, fewer security features, and less flexibility.	More complex than traditional BIOS and may have compatibility issues with very old operating systems or hardware.
Modern Usage	Mostly found in older computers and legacy configurations.	The standard choice for modern computers and operating systems. Microsoft recommends using UEFI mode when possible because it provides additional security features. (Microsoft Learn)



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Why UEFI Has Largely Replaced BIOS

Both BIOS and UEFI perform an important function in a computer because they provide the firmware environment that operates when the computer is started. However, BIOS (Basic Input/Output System) is an older technology that was designed for the hardware and storage requirements of earlier generations of computers. As computers became more powerful and storage devices became larger, the limitations of BIOS became more noticeable. UEFI (Unified Extensible Firmware Interface) was developed as a replacement for the older BIOS firmware interface and provides a more modern environment for starting and managing an operating system.

One of the major improvements of UEFI is its support for GPT (GUID Partition Table). Traditional BIOS systems commonly use MBR, which has limitations on disk capacity and partition management. GPT uses 64-bit logical block addresses and supports many more partitions than the traditional MBR structure. It also stores backup partition information and uses CRC32 checks for improved data integrity. This makes GPT more suitable for modern computers with large storage devices. UEFI also provides important security improvements. One of its most important features is Secure Boot. When Secure Boot is enabled, the firmware checks the digital signature of boot software before allowing it to run. This helps prevent unauthorized or malicious software from being loaded during the startup process. Modern Windows security features such as Secure Boot and Measured Boot depend on UEFI-based systems.

Another advantage is booting performance. Microsoft identifies faster boot and resume times as one of the benefits of UEFI. UEFI also provides a more flexible boot environment because it can load UEFI drivers and applications and can manage boot options through its firmware environment.

From a system administration perspective, UEFI is also more practical for modern enterprise computers. System administrators commonly work with large storage devices, multiple partitions, modern operating systems, and security requirements. UEFI and GPT provide better support for these requirements than the older BIOS and MBR combination. Microsoft documentation also recommends installing Windows using UEFI mode, when possible, because of its additional security advantages.

In conclusion, UEFI has largely replaced BIOS because it is better suited to modern computer hardware and operating systems. BIOS remains useful when maintaining older computers or legacy systems, but its limitations in storage capacity, partitioning, security, and flexibility make it less appropriate for newer computers. UEFI provides improved boot performance, support for GPT and large storage devices, and security features such as Secure Boot. Therefore, understanding the difference between BIOS and UEFI is important for system administrators because firmware configuration can affect operating-system installation, disk partitioning, booting, and system security.



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References

- [Microsoft Learn – UEFI firmware requirements](#)
- [Microsoft Learn – Boot to UEFI Mode or Legacy BIOS Mode](#)
- [Microsoft Learn – Windows support for hard disks exceeding 2 TB](#)
- [UEFI Forum – UEFI Specification](#)
- [Microsoft Support – Windows 11 and Secure Boot](#)